Technical Coding Challenge: AI Loan Adjudication.

Role: AI/ML Technology Lead
Objective: Design, implement, and evaluate a process for loan adjudication application.

Challenge Details:

1. **Dataset**:
    - A sample dataset of credit loan rules is provided in a JSON format. Each entry contains a rule that must be satisfied and followed.
2. **Tasks**:
    - **Data Preparation**:
        - Write a Python script to:
            - Load the provided JSON dataset.
            - Preprocess the dialogues into a suitable format for LLM training. This may involve:
                - Tokenization. (Special token addition, Padding or truncation)
            - Split the data into training and validation sets.
            - Discuss your rationale for the chosen preprocessing steps.
    - **Model Selection and Training**:
        - Free to choose any ML-Model for the task.
        - You can also use a pre-trained LLM from Hugging Face Transformers as a base model.
    - **Evaluation and Analysis**:
        - Evaluate your model's performance on the validation set.
        - Generate sample applications using your model and analyze the model's strengths and weaknesses.
        - The output should be adjudication decision, explaining the user rationale behind the decision.

3. **Deliverables**:
    - A GitHub repository containing:
        - Python scripts for data preparation, model training, and evaluation.
        - A README file with detailed instructions.
        - (Optional) A brief report (1-2 pages) summarizing the approach, results, and analysis.